CPE412 Pattern Recognition

Week 12

Principal Component Analysis (PCA)

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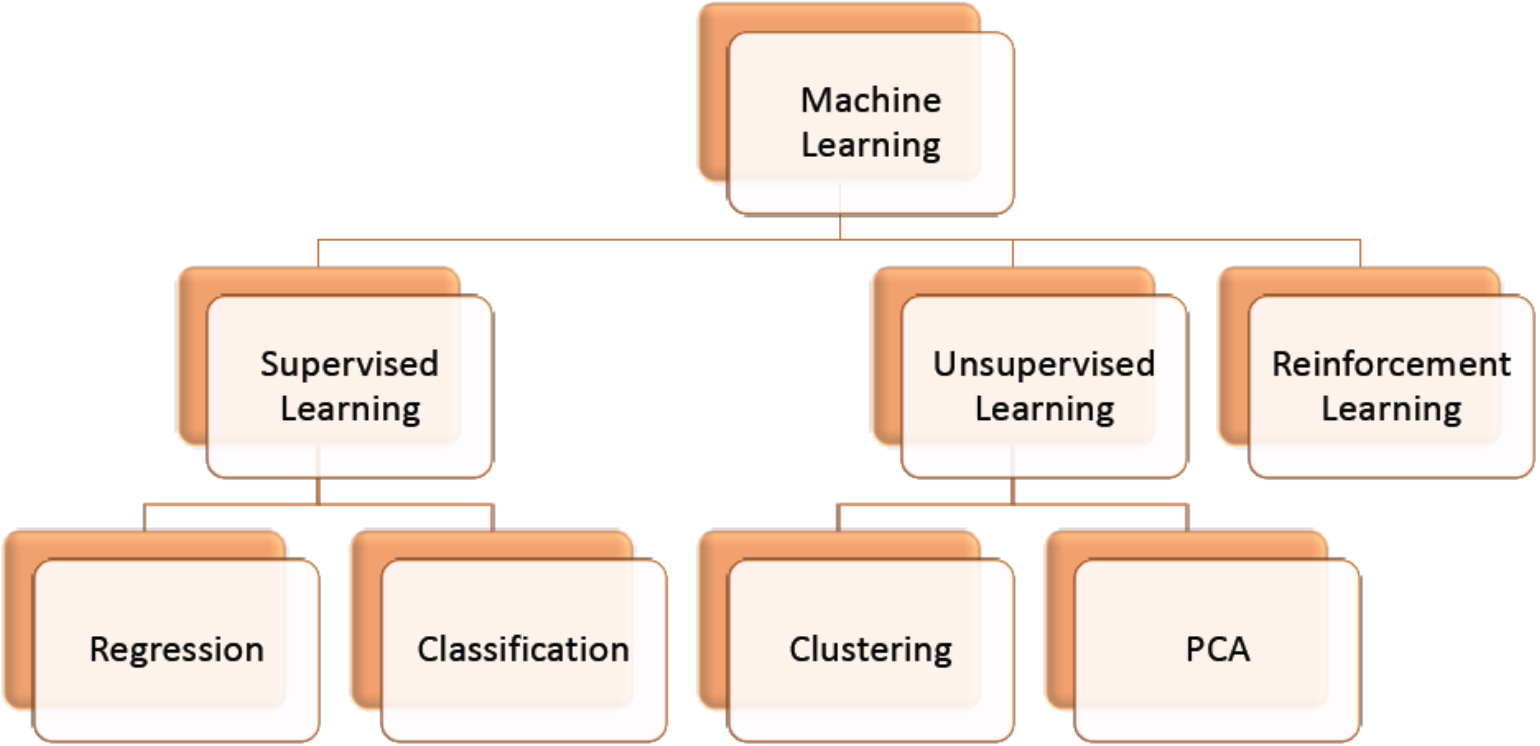
Types of Learning



Principal Component Analysis(PCA)| Dimension Reduction

- PCA is an unsupervised learning algorithm that is used for the dimensionality reduction in machine learning.
- It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features. These new transformed features are called the Principal Components.
- It is one of the popular tools that is used for exploratory data analysis and predictive modeling.
- It is a technique to draw strong patterns from the given dataset by reducing the variances.
- PCA generally tries to find the lower-dimensional surface to project the high-dimensional data.

Dimension Reduction

- In pattern recognition, Dimension Reduction is defined as-
  - It is a process of converting a data set having vast dimensions into a data set with lesser dimensions.
  - It ensures that the converted data set conveys similar information concisely.

PCA Applications

- Data Visualization
- Data Compression
- Noise Reduction
- Data Classification
- …

Data Visualization

- Example:
  - Given 53 blood and urine measurements(features) from 65 individuals
  - How can we visualize the measurements?

Data Visualization

- Matrix format (65x53)

Instances

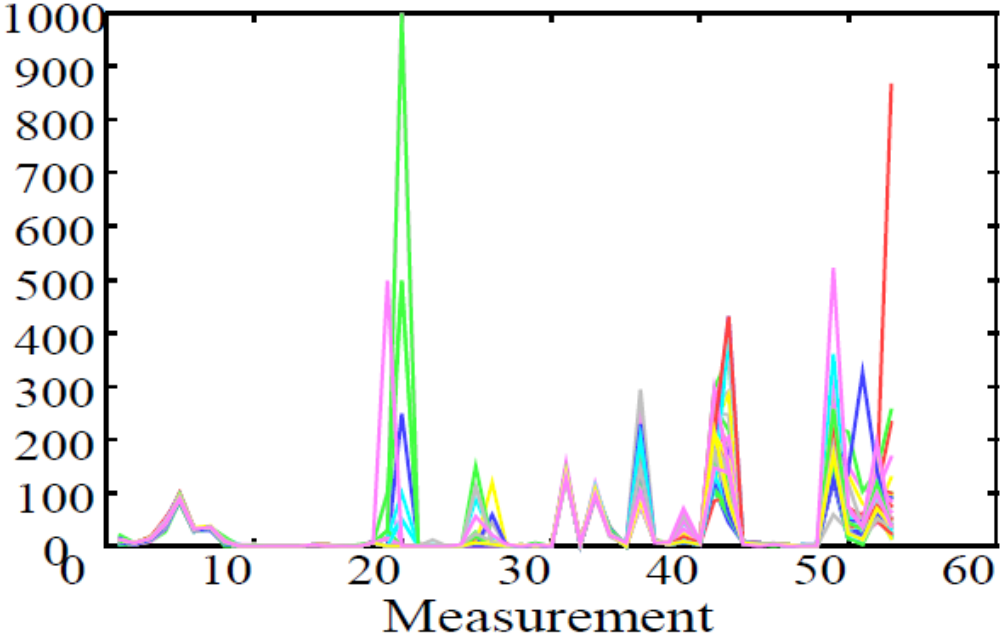
	H-WBC	H-RBC	H-Hgb	H-Hct	H-MCV	H-MCH	H-MCHC
A1	8.0000	4.8200	14.1000	41.0000	85.0000	29.0000	34.0000
A2	7.3000	5.0200	14.7000	43.0000	86.0000	29.0000	34.0000
A3	4.3000	4.4800	14.1000	41.0000	91.0000	32.0000	35.0000
A4	7.5000	4.4700	14.9000	45.0000	101.0000	33.0000	33.0000
A5	7.3000	5.5200	15.4000	46.0000	84.0000	28.0000	33.0000
A6	6.9000	4.8600	16.0000	47.0000	97.0000	33.0000	34.0000
A7	7.8000	4.6800	14.7000	43.0000	92.0000	31.0000	34.0000
A8	8.6000	4.8200	15.8000	42.0000	88.0000	33.0000	37.0000
A9	5.1000	4.7100	14.0000	43.0000	92.0000	30.0000	32.0000

Features

Difficult to see the correlations between the features...

Data Visualization

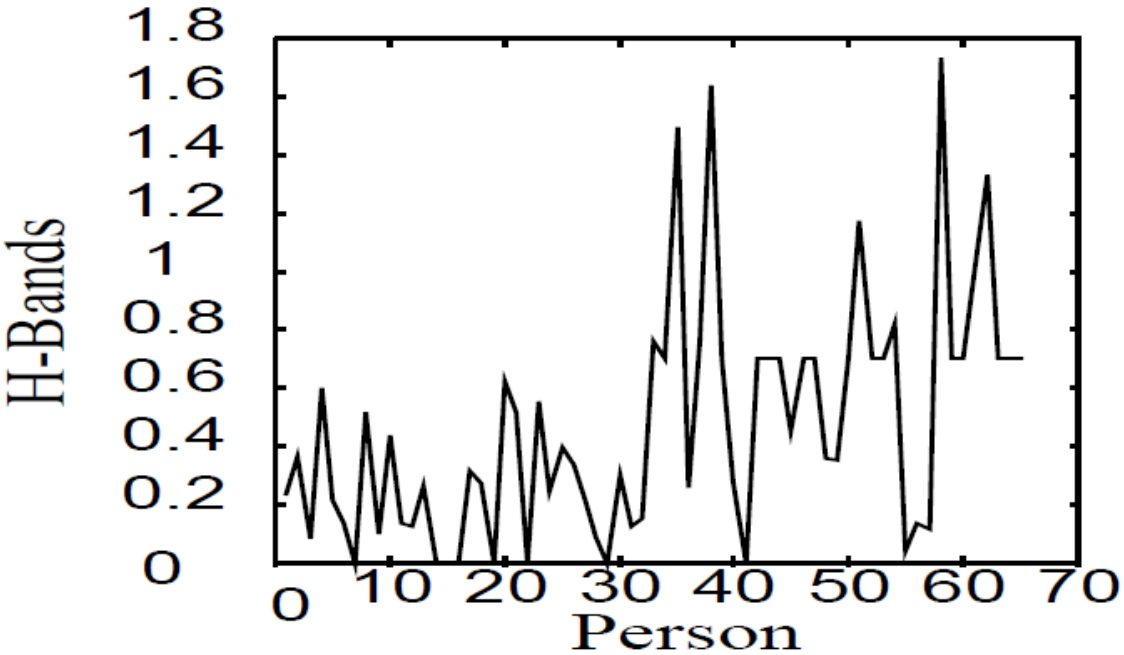
- Spectral format (65 pictures, one for each person)



Difficult to compare the different patients…

Data Visualization

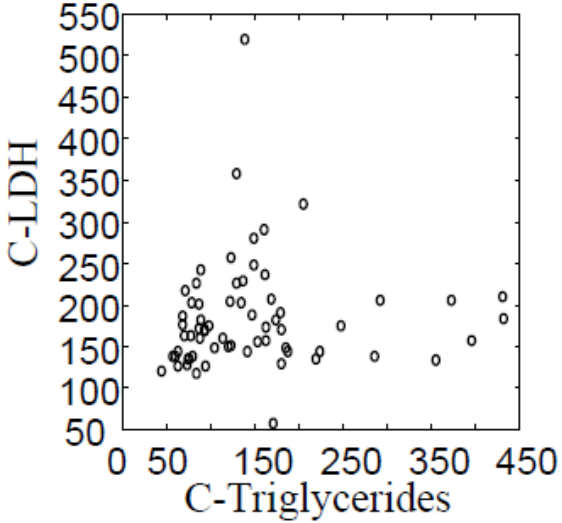
- Spectral format (53 pictures, one for each feature)



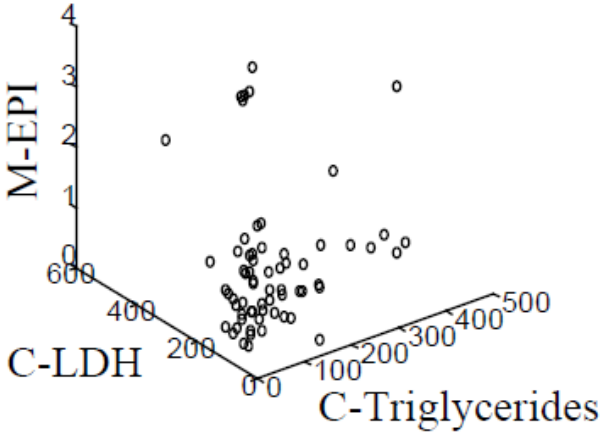
Difficult to see the correlations between the features...

Data Visualization

Bi-variate



Tri-variate



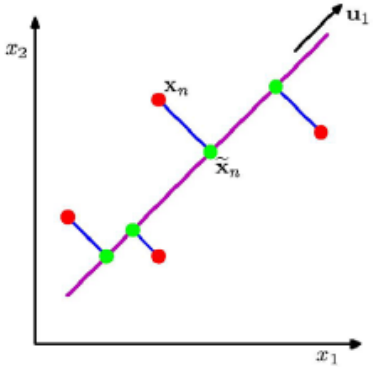
How can we visualize the other variables???

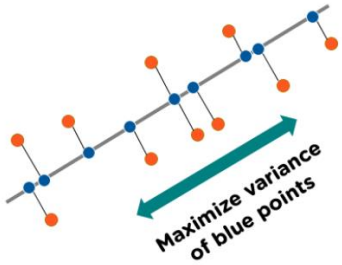
... difficult to see in 4 or higher dimensional spaces...

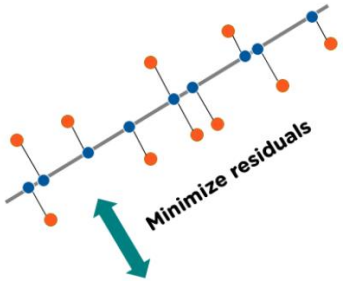
Data Visualization

- Is there a better representation than the coordinate axes?
- Is it really necessary to show all the 53 dimensions?
- … what if there are strong correlations between some of the features?
- How could we find the smallest subspace of the 53-D space that keeps the most information about the original data?
- A solution: Principal Component Analysis

Principal Component Analysis







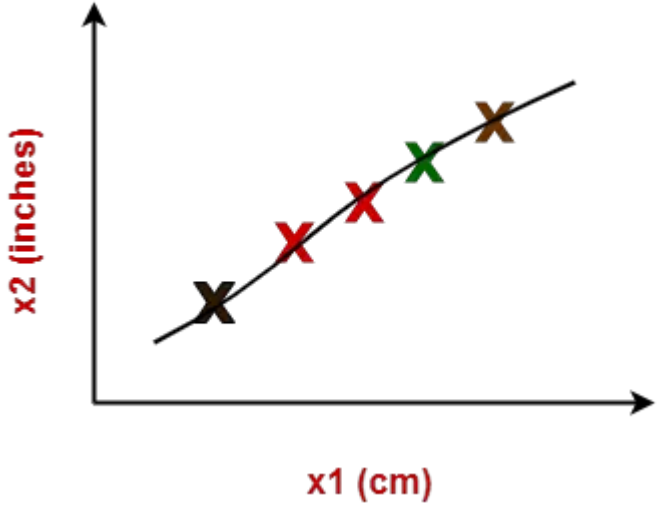
PCA:

Orthogonal projection of data onto lower-dimension linear space that...

- maximizes variance of projected data (purple line)
- minimizes mean squared distance between data points and their projections (the blue segments)

Example

- The following graph shows two dimensions x1 and x2.
- x1 represents the measurement of several objects in cm.
- x2 represents the measurement of several objects in inches.



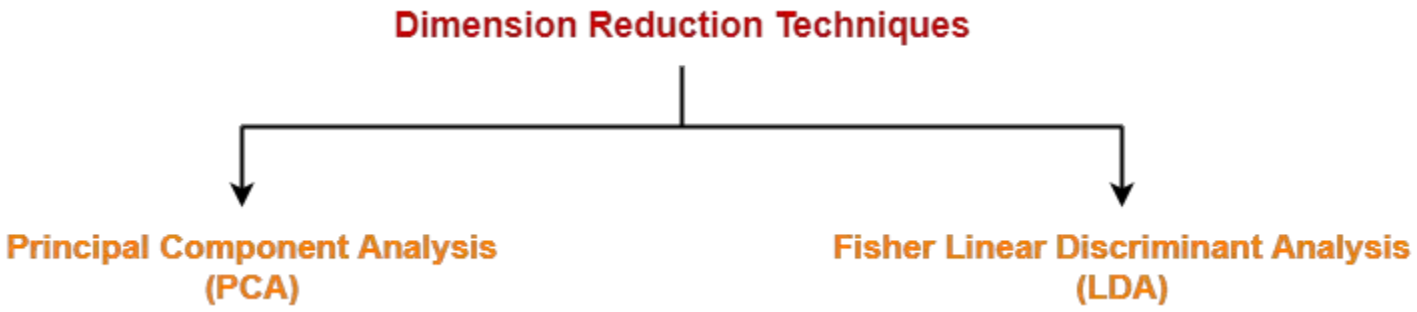
Example

- In machine learning,
- Using both these dimensions convey similar information.
- Also, they introduce a lot of noise in the system.
- So, it is better to use just one dimension.
- Using dimension reduction techniques–
  - We convert the dimensions of data from 2 dimensions (x1 and x2) to 1 dimension (z1).
  - It makes the data relatively easier to explain.

Benefits–

- It compresses the data and thus reduces the storage space requirements.
- It reduces the time required for computation since less dimensions require less computation.
- It eliminates the redundant features.
- It improves the model performance.
- Overfitting Prevention: High dimensional data may lead to overfitting in machine learning models

Dimension Reduction Techniques-



PCA Algorithm

- Step-01: Get data.
- Step-02: Compute the mean vector (μ)(in some references $\bar{\mathbf{x}}$).
- Step-03: Subtract mean from the given data.
- Step-04: Calculate the covariance matrix.
- Step-05: Calculate the eigen vectors and eigen values of the covariance matrix.
- Step-06: Choosing components and forming a feature vector.
- Step-07: Deriving the new data set.

PCA-covariance matrix

μ

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\text{Covariance Matrix} = \frac{\Sigma (x_i - \mu)(x_i - \mu)^t}{n}$$

covariance is a measure of the relationship between two random variables.

Problem-01:

- Given data = { 2, 3, 4, 5, 6, 7 ; 1, 5, 3, 6, 7, 8 }.
- Compute the principal component using PCA Algorithm.
- **OR**
  - Compute the principal component of following data-

CLASS 1

X = 2 , 3 , 4

Y = 1 , 5 , 3

CLASS 2

X = 5 , 6 , 7

Y = 6 , 7 , 8

X	Y
2	1
3	5
4	3
5	6
6	7
7	8

Problem-01:Solution

- Step-01: Get data.
- The given feature vectors are-
- x1 = (2, 1)
- x2 = (3, 5)
- x3 = (4, 3)
- x4 = (5, 6)
- x5 = (6, 7)
- x6 = (7, 8)

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \begin{bmatrix} 6 \\ 7 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Problem-01:Solution

- Step-02: Calculate the mean vector (μ).
- Mean vector (μ)
- = ((2 + 3 + 4 + 5 + 6 + 7) / 6, (1 + 5 + 3 + 6 + 7 + 8) / 6)
- = (4.5, 5)
- Thus,

$$\text{Mean vector } (\mu) = \begin{bmatrix} 4.5 \\ 5 \end{bmatrix}$$

Problem-01:Solution

- Step-03:Subtract mean vector (μ) from the given feature vectors.
  - $x1 - \mu = (2 - 4.5, 1 - 5) = (-2.5, -4)$
  - $x2 - \mu = (3 - 4.5, 5 - 5) = (-1.5, 0)$
  - $x3 - \mu = (4 - 4.5, 3 - 5) = (-0.5, -2)$
  - $x4 - \mu = (5 - 4.5, 6 - 5) = (0.5, 1)$
  - $x5 - \mu = (6 - 4.5, 7 - 5) = (1.5, 2)$
  - $x6 - \mu = (7 - 4.5, 8 - 5) = (2.5, 3)$
- Feature vectors (xi) after subtracting mean vector (μ) are-

$$\begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 2.5 \\ 3 \end{bmatrix}$$

Problem-01:Solution

- Step-04:Calculate the covariance matrix.
- Covariance matrix is given by-

$$\text{Covariance Matrix} = \frac{\Sigma (x_i - \mu)(x_i - \mu)^t}{n}$$

Problem-01:Solution

$$m_1 = (x_1 - \mu)(x_1 - \mu)^t = \begin{bmatrix} -2.5 \\ -4 \end{bmatrix} \begin{bmatrix} -2.5 & -4 \end{bmatrix} = \begin{bmatrix} 6.25 & 10 \\ 10 & 16 \end{bmatrix}$$

$$m_2 = (x_2 - \mu)(x_2 - \mu)^t = \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \begin{bmatrix} -1.5 & 0 \end{bmatrix} = \begin{bmatrix} 2.25 & 0 \\ 0 & 0 \end{bmatrix}$$

$$m_3 = (x_3 - \mu)(x_3 - \mu)^t = \begin{bmatrix} -0.5 \\ -2 \end{bmatrix} \begin{bmatrix} -0.5 & -2 \end{bmatrix} = \begin{bmatrix} 0.25 & 1 \\ 1 & 4 \end{bmatrix}$$

$$m_4 = (x_4 - \mu)(x_4 - \mu)^t = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$m_5 = (x_5 - \mu)(x_5 - \mu)^t = \begin{bmatrix} 1.5 \\ 2 \end{bmatrix} \begin{bmatrix} 1.5 & 2 \end{bmatrix} = \begin{bmatrix} 2.25 & 3 \\ 3 & 4 \end{bmatrix}$$

$$m_6 = (x_6 - \mu)(x_6 - \mu)^t = \begin{bmatrix} 2.5 \\ 3 \end{bmatrix} \begin{bmatrix} 2.5 & 3 \end{bmatrix} = \begin{bmatrix} 6.25 & 7.5 \\ 7.5 & 9 \end{bmatrix}$$

Problem-01: Solution

- Now, Covariance matrix
- = (m1 + m2 + m3 + m4 + m5 + m6) / 6
- On adding the above matrices and dividing by 6, we get-

$$\text{Covariance Matrix} = \frac{1}{6}\begin{bmatrix} 17.5 & 22 \\ 22 & 34 \end{bmatrix}$$

$$\text{Covariance Matrix} = \begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix}$$

Problem-01: Solution

- Step-05: Calculate the eigen values and eigen vectors of the covariance matrix.
- λ is an eigen value for a matrix M if it is a solution of the characteristic equation $|M - \lambda| = 0$.
- So, we have-

$$\begin{vmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2.92 - \lambda & 3.67 \\ 3.67 & 5.67 - \lambda \end{vmatrix} = 0$$

Problem-01: Solution

- From here,
- $(2.92 - \lambda)(5.67 - \lambda) - (3.67 \times 3.67) = 0$
- $16.56 - 2.92\lambda - 5.67\lambda + \lambda^2 - 13.47 = 0$
- $\lambda^2 - 8.59\lambda + 3.09 = 0$
- Solving this quadratic equation, we get $\lambda = 8.22, 0.38$
- Thus, two eigen values are $\lambda 1 = 8.22$ and $\lambda 2 = 0.38$.
- Clearly, the second eigen value is very small compared to the first eigen value. So, the second eigen vector can be left out.

Problem-01: Solution

- Eigen vector corresponding to the greatest eigen value is the principal component for the given data set.
- So. we find the eigen vector corresponding to eigen value $\lambda 1$.
- We use the following equation to find the eigen vector-
  - $MX = \lambda X$

Problem-01: Solution

- Eigen vector corresponding to the greatest eigen value is the principal component for the given data set.
- So. we find the eigen vector corresponding to eigen value $\lambda 1$.
- We use the following equation to find the eigen vector-
  - $MX = \lambda X$
  - where-
  - M = Covariance Matrix
  - X = Eigen vector
  - λ = Eigen value

Problem-01: Solution

- Substituting the values in the above equation, we get-

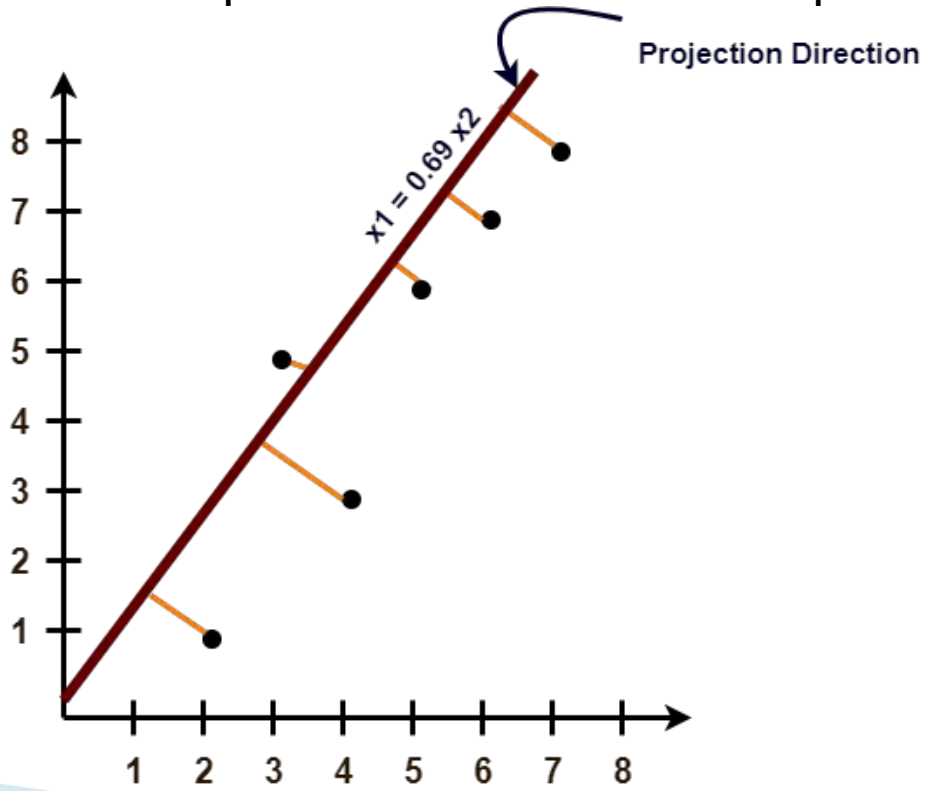
$$\begin{bmatrix} 2.92 & 3.67 \\ 3.67 & 5.67 \end{bmatrix} \begin{bmatrix} X1 \\ X2 \end{bmatrix} = 8.22 \begin{bmatrix} X1 \\ X2 \end{bmatrix}$$

- Solving these, we get-
  - $2.92X1 + 3.67X2 = 8.22X1$
  - $3.67X1 + 5.67X2 = 8.22X2$
- On simplification, we get-
  - $5.3X1 = 3.67X2$ .....(1)
  - $3.67X1 = 2.55X2$ .....(2)
- From (1) and (2), $X1 = 0.69X2$
- From (2), the eigen vector is

Eigen Vector : $\begin{bmatrix} X1 \\ X2 \end{bmatrix} = \begin{bmatrix} 2.55 \\ 3.67 \end{bmatrix}$

Problem-01: Solution

- Lastly, we project the data points onto the new subspace as-



Try it yourself

x	y
2,5	2,4
0,5	0,7
2,2	2,9
1,9	2,2
3,1	3
2,3	2,7
2	1,6
1	1,1
1,5	1,6
1,1	0,9

After this slides just a help for solving equations

Perfect square equation

perfect squares

$$a^2 + 2ab + b^2 = (a+b)(a+b)$$

$$a^2 - 2ab + b^2 = (a-b)(a-b)$$

difference of squares

$$a^2 - b^2 = (a+b)(a-b)$$

Factoring perfect square trinomials

$$x^2 + 6x + 9$$

$$x^2 + 6x + 3^2$$

$$(x + 3)^2$$

$$(x + 3)(x + 3)$$

Solution of quadratic equation

$r^2 - r - 2 = 0$

Solution of quadratic equation($ax^2 + bx + c = 0$)is given by -

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \ldots (1)$$

Here, a = 1, b = 1 and c = 1 ($x^2 + x + 1 = 0$)

Just put the value of a, b and c in eq(1),

$$x = \frac{-1 \pm \sqrt{1^2 - 4*1*1}}{2*1}$$

$$x = \frac{-1 \pm \sqrt{-3}}{2}$$

Thanks ☺